Daniel Cohen

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Google Scholar

Summary

Applied Scientist and Research Scientist with 10+ years in Information Retrieval, NLP, and Machine Learning. Ph.D., Computer Science, UMass Amherst. Currently building production LLM and agentic AI systems at Dataminr processing 200M+ real-time events per day. Two best-paper awards; consistent publication record at ACL, NAACL, SIGIR, EMNLP, CIKM, and ICML.

Technical Skills

- **Programming:** Python, PyTorch, TensorFlow, JAX, CUDA, SQL, Bash, Java, C
- **ML & AI:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, Generative AI, Neural Ranking, Information Retrieval, NLP, RLHF, Reinforcement Learning, Adversarial Learning, Uncertainty Quantification, Calibration, Fairness & Bias Mitigation, Online Learning
- **Frameworks & Tools:** Hugging Face Transformers, Scikit-learn, Numpy, Scipy, Pandas, Galago, PyLucene, Linux/Unix, git

Experience

- **Research Scientist** – Dataminr — Aug. 2022 – present
  - Designed and deployed production generative ML and LLM systems (ReGenAI) processing 200M+ real-time events per day, enabling automated event brief generation and critical intelligence delivery at scale.
  - Developed domain-specialized LLMs and multi-agent agentic AI architectures (Intel Agents) that autonomously reason over high-volume real-time data streams to surface critical events, threats, and risks.
  - Researched model stability, calibration, and robustness under distribution shift for high-throughput online learning systems operating in adversarial and noisy streaming environments.
- **Postdoctoral Researcher** – Brown University (Prof. Carsten Eickhoff) — Dec. 2020 – July 2022
  - Executed an independent research agenda on uncertainty-aware and fairness-constrained ranking; mentored PhD and undergraduate researchers and delivered all grant milestones on schedule.
- **Research Assistant** – UMass Amherst (Prof. W. Bruce Croft) — Sep. 2015 – Dec. 2020
  - Developed novel training methods and robustness techniques for neural IR and QA models, yielding publications at SIGIR, EMNLP, ICTIR, ICML, and ECIR and two best-paper awards.
- **Site Project Manager** – UMass Amherst (IARPA MATERIAL) — Sep. 2017 – Dec. 2020
  - Led UMass Amherst deliverables for a multi-university IARPA grant; coordinated biweekly reporting and annual cross-lingual IR system evaluations, achieving the top retrieval scores across all competing teams.
- **Research Intern** – Microsoft Research (Dr. Fernando Diaz, Dr. Bhaskar Mitra) — Feb. 2019 – Jun. 2019
  - Applied dense retrieval techniques to build a scalable policy index for task transfer in reinforcement learning, enabling efficient policy lookup without expensive pairwise comparisons or simultaneous multi-task training.
- **Research Intern** – Microsoft Research (Dr. Katja Hofmann, Dr. Bhaskar Mitra) — May 2017 – Aug. 2017
  - Developed an adversarial domain adaptation method for neural ranking models, producing a robust, domain-agnostic retrieval system that generalizes to out-of-distribution markets with minimal labeled data (SIGIR 2018 Best Short Paper).
- **Research Assistant** – New York University (Prof. Mohamed Zahran) — Jun. 2014 – Jan. 2015
  - Built a GPU performance profiling toolkit for CUDA code using a simulated GPU backend, enabling architecture-agnostic identification of processor and memory bottlenecks.

Education

- Ph.D. **Computer Science**. University of Massachusetts Amherst. — 2015 – 2020
- M.S. **Computer Science**. University of Massachusetts Amherst. — 2015 – 2017
- B.A. **Computer Science and Mathematics**. New York University. — 2013 – 2015

Selected Publications

- **Explain then Rank: Scale Calibration of Neural Rankers Using Natural Language Explanations from LLMs.** P. Yu, **D. Cohen**, H. Lamba, J. Tetreault, A. Jaimes. *ACL (Findings)*, 2025.
- **In-Context Example Ordering Guided by Label Distributions.** Z. Xu, **D. Cohen**, B. Wang, V. Srikumar. *NAACL (Findings)*, 2024.
- **Predictive Uncertainty-based Bias Mitigation in Ranking.** M. Hauss, **D. Cohen**, M. Mansoury, M. de Rijke, C. Eickhoff. *CIKM*, 2023.
- **A Lightweight Constrained Generation Alternative for Query-Focused Summarization.** Z. Xu, **D. Cohen**. *SIGIR*, 2023.
- **Evaluating the Performance of Reinforcement Learning Algorithms.** S. Jordan, Y. Chandak, **D. Cohen**, M. Zhang, P. Thomas. *ICML*, 2020.
- **Learning a Better Negative Sampling Policy with Deep Neural Networks for Search. D. Cohen**, S. Jordan, W.B. Croft. *ICTIR*, 2019. **Best Full Paper Award.**
- **Cross Domain Regularization for Neural Ranking Models Using Adversarial Learning. D. Cohen**, B. Mitra, K. Hofmann, W.B. Croft. *SIGIR*, 2018. **Best Short Paper Award.**

Full list: Google Scholar · dscohen.github.io/publications

Awards and Grants

- IARPA BETTER Research Grant, co-writer. 2020
- Best Full Paper, ICTIR 2019
- SIGIR Student Travel Grant 2019
- Bloomberg Data Science Research Grant, co-writer. 2018
- Best Short Paper, SIGIR 2018
- SIGIR Student Travel Grant 2018
- IARPA MATERIAL Research Grant, co-writer. 2017
- SIGIR Student Travel Grant 2016
- Best Poster – NYU Undergraduate Research Conference 2016

Professional Activities and Service

- **Program Committee Member:** SIGIR, EMNLP, EACL, CIKM, WWW, WSDM, ACL, KDD, AAAI, TOIS, TACL